

# Alessandro Linzi

**PhD Mathematician | AI & Machine Learning | Bridging Mathematical Foundations with Computational Intelligence**

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## Core Competencies

**Mathematical Areas:** Model Theory, Valued Fields, Hypercompositional Algebra, Mathematical Logic, Category Theory

**Technical Skills:** Python programming, Proof assistants (Isabelle, Agda, Lean, Coq), LaTeX, Mathematical computing

**Research Experience:** Independent research design, International collaboration, Peer review, Editorial work

**Communication:** Academic writing, Conference presentations, Teaching, Multilingual (Italian, English, Spanish)

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## Recent Training

### Profession AI | September 2025

**AI avanzata (Summer Camp 2025)** - 7-hour advanced AI course

Certificate ID: 68cbe06459039b762f09e80d

Instructor: Gianluca Malato, Head of Teaching

### Profession AI | September 2025

**Programmazione con Python (Summer Camp 2025)** - 15-hour Python programming course

Certificate ID: 68b80a1c77a97ff7840e6904

Instructor: Gianluca Malato, Head of Teaching

### Anthropic AI | October 2025

**AI Fluency: Framework & Foundations** - Foundational AI competency certification

Issuing organizations: Anthropic, University College Cork (Ireland), Ringling College of Art + Design

Verification URL: <http://verify.skilljar.com/c/byokkh2frv9y>

**Anthropic AI | October 2025**

**AI Fluency for Educators** - Comprehensive AI literacy certification program

Issuing organizations: Anthropic, University College Cork (Ireland), Ringling College of Art + Design

Verification URL: <http://verify.skilljar.com/c/qjkhvhuwe7ina>

**Anthropic AI | October 2025**

**Teaching the AI Fluency Framework** - Advanced pedagogical training for AI education

Issuing organizations: Anthropic, University College Cork (Ireland), Ringling College of Art + Design

Verification URL: <http://verify.skilljar.com/c/ew3ajgipuktw>

**Anthropic AI | October 2025**

**Claude Code in Action** - Hands-on training for AI-assisted software development and coding workflows

Issuing organizations: Anthropic

Verification URL: <http://verify.skilljar.com/c/eqthnycdhn3i>

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**Professional Experience****ISIS E. Mattei di Latisana, Italy****Mathematics Teacher | 2025**

- Teaching mathematics to 6 classes across high school years 1-4
- Maintaining documentation systems and curriculum planning

**University of Nova Gorica, Slovenia****Assistant Professor (Postdoc) | 2022–2024**

Centre for Information Technology and Applied Mathematics

- Conducted research resulting in 8 peer-reviewed publications and 5 submissions
- Achieved habilitation as Assistant in Mathematics in Slovenia
- Served as editor and peer reviewer for mathematical journals
- Participated in promotion activities for School of Engineering and Management
- Co-authored textbook: “Mathematics for Engineering and Management” (ISBN: 978-961-7025-41-5)

**University of Szczecin, Poland****PhD Researcher & Teaching Assistant | 2018–2022**

Department of Mathematics

- Published 2 peer-reviewed papers during doctoral studies
- Taught undergraduate course “Information Technologies” for School of Humanities

- Assisted in mentoring master's students
- Participated in international valuation theory research group

### **Universities of Florence & Udine, Italy**

#### **Teaching Assistant & Tutor | 2016–2018**

- Mathematics tutoring for School of Agriculture students
- Developed teaching materials for diverse student backgrounds

#### **Internship – AFS Intercultura (Colle Val d'Elsa, SI, Italy)**

**Mar 2014 – Oct 2014**

- Supported the organization of the student hosting program using Intercultura's management tools.
- Reviewed and evaluated profiles of foreign students applying to exchange programs.
- Prepared summary reports and communicated results to local coordinators.
- Entered and updated student data in digital reports, databases, and tools such as Global Link and Filemaker.
- Participated in training meetings with and for foreign students.
- Managed email correspondence in both Italian and English; supported logistics for student departures with host families.

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## **Education**

### **University of Szczecin, Poland**

#### **PhD in Mathematics | 2018–2022**

Dissertation: “Algebraic hyperstructures in the model theory of valued fields”

Supervisor: Prof. Franz-Viktor Kuhlmann

Funding: EU scholarship (4,000 PLN/month), Outstanding student salary compensation

### **University of Florence, Italy**

#### **Master's in Mathematics (Didactical Curriculum), 110/110 Cum Laude | 2016–2018**

Thesis: “The Theorem of Ax-Kochen-Ershov”

Supervisor: Prof. Antongiulio Fornasiero

Award: University prize for exceptional performance

### **University of Udine, Italy**

#### **Bachelor's in Mathematics, 101/110 | 2010–2015**

Thesis: “The Consistency of Arithmetics”

Supervisor: Prof. Franco Parlamento

## AFS Costa Rica

### International Exchange Program | 2008–2009

6-month intercultural immersion program during high school

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## Technical Skills

**Programming:** Python (recent certification), LaTeX, Git

**Mathematical Software:** Isabelle, Agda, MATLAB, Excel

**Research Tools:** Citation management, collaborative platforms, web technologies

**AI/ML Familiarity:** Experience with language models, interest in automated theorem proving

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## Publications

### Peer-reviewed Articles:

1. P. Blaszkiewicz, H. Cmiel, **A. Linzi**, P. Szewczyk, “Oettli-Théra and Caristi-Kirk ball spaces and applications”, *J. of fixed point theory and applications*, Vol. 21, No. 98 (2019).
2. K. Kuhlmann, **A. Linzi**, H. Stojalowska, “Orderings and valuations in hyperfields”, *J. of Algebra*, Vol. 611 (2022), pp. 399-421.
3. D. E. Kędzierski, **A. Linzi** and H. Stojalowska, “Characteristic, C-characteristic and positive cones in hyperfields”, *Mathematics*, 11(3): 779 (2023).
4. **A. Linzi** and I. Cristea, “Dependence relations and grade fuzzy set”, *Symmetry*, 15(2):311 (2023).
5. **A. Linzi**, “A result of Krasner in categorial form”, *Mathematics*, 11(24): 4923 (2023).
6. **A. Linzi**, “Polygroup objects in regular categories”, *AIMS Mathematics*, 9(5) (2024).
7. **A. Linzi**, “Notes on valuation theory for Krasner hyperfields”, *Isr. J. Math.* (2025).
8. **A. Linzi** and I. Cristea, “Preface to the Special Issue”Algebraic Structures and Graph Theory, 2nd Edition”“, *Mathematics*, 13(4):577 (2025).
9. **A. Linzi** and P. Touchard, “On the hyperfields associated to valued fields”, *Journal of Pure and Applied Algebra* (2025).

10. **A. Linzi**, H. Stojalowska, “Hypervaluations on hyperfields and ordered canonical hypergroups”, *Iranian J. of mathematical sciences and informatics* (2025).

**Accepted articles:**

1. **A. Linzi**. On algebraically closed Krasner hyperfields. accepted by the *Italian Journal of Pure and Applied Mathematics*.
  2. N. Cangiotti, **A. Linzi** and E. Talotti. L-mosaics and orthomodular lattices. accepted by the *Italian Journal of Pure and Applied Mathematics*.
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## Professional Service

**Editorial Work:** Special Issue Editor, MDPI Mathematics (2025) - “Algebraic Structures and Graph Theory”

**Peer Review:** Completed 13 reviews for journals including Journal of Algebra, Italian Journal of Pure and Applied Mathematics, Heliyon

**Conference Participation:** Regular presenter at international mathematics conferences across Europe

**International Programs:** Erasmus+ Teaching Program participant (Montenegro, 2023-2025)

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## Selected Conference Presentations

- The 4th International Conference on Symmetry, Barcelona, Spain (2023)
  - 20th International Conference on Quantum Physics and Logic (QPL 2023), Paris, France (2023)
  - Logic4Peace Conference, Amsterdam, Netherlands (2022)
  - Symposium on Hypercompositional Algebra, Nova Gorica, Slovenia (2022)
  - Multiple invited seminars across European universities (2019–2024)
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## Awards & Scholarships

- EU Co-funded PhD Scholarship with full tuition and monthly stipend (2018–2022)
- Outstanding Student Salary Compensation, University of Szczecin (2018–2021)
- University of Florence Prize for Exceptional Academic Performance (2018)

- AFS Intercultural Exchange Scholarship (2008)
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## Languages

**Italian:** Native speaker

**English:** Fluent (academic and professional)

**Spanish:** Fluent

**Slovenian:** Basic

**Polish:** Basic

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## Research Interests

- Foundations of mathematics and mathematical logic
- Hypercompositional algebra and valued field theory
- Applications of AI and machine learning to mathematical research
- Proof assistants and automated theorem proving
- Constructive logic and type theory